

TECHNICAL AND TACTICAL CHALLENGE OF TRAINING IN AUSTRALIAN FOOTBALL

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INTRODUCTION

Coaches design training to be optimally challenging for the athletes to yield learning and transfer to competition. The Challenge Point Framework (CPF) (1) proposes that the optimal challenge point is a product of the interaction of task difficulty and the learner's skill level. The Rating of Perceived Challenge (RPC) (2) allows coaches to quantify perceived challenge of training and has been shown to be a valid tool when implemented in Australian Football (3). Despite its validity, the RPC only provides a global rating of challenge, not offering insight into the specific technical and tactical components of challenge within training, which limits its use by coaches (2).

This study examined whether technical and tactical RPC scores differ between training tasks (Aim 1), and the differences in objective performance indicators across different levels of perceived technical and tactical challenge (Aim 2).

METHODS

Twenty-two elite Australian Football players completed a modified RPC across the preseason and early 2026 season. Thirty-five drills were classified as skills or game-sense training drills. After selected drills, players rated their perceived technical and tactical challenge using an adapted RPC scale (2). Furthermore, objective performance indicators in the drills were coded from video using SportsCode. For Aim 1, a linear mixed-effects model tested drill type, RPC type and their interaction, with player and session as random effects. For Aim 2, player-level data (i.e., individual RPC scores and objective performance indicators) were analysed using MANOVA's on game-sense and skills drills with significant multivariate effects followed-up with univariate analyses.

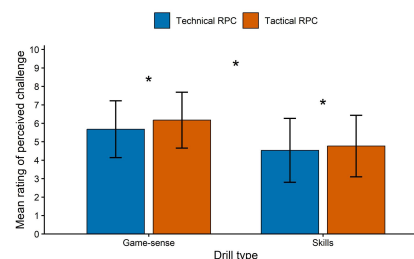
RESULTS AND DISCUSSION

Game-sense training produced higher RPC scores compared to skills training (Skills vs Game-sense: $b = -1.159$, 95% CI $[-1.630, -0.688]$, $t(26.7) = -4.84$, $p < .001$), and tactical RPC was higher overall than technical RPC ($b = 0.359$, 95% CI $[0.093, 0.626]$, $t(330.7) = 2.65$, $p = .008$; Table 1). The drill type $\times$ RPC type interaction was not significant ($p = .334$), providing no evidence that the technical-tactical difference varied between the two drill types. The model explained 43.6% of differences when fixed and random effects were considered. For Aim 2, only skills training tactical RPC showed a significant multivariate effect (Pillai's $V = .897$, $F(36, 272) = 2.18$, $p < .001$), that had a significant univariate

effect for disposal efficiency. Disposal efficiency differed across tactical RPC levels ($F(4, 73) = 4.19$, $p = .004$).

Figure 1.

Mean Technical and Tactical Rating of Perceived Challenge Scores by Drill Type



Note: Bars represent observed mean RPC scores, and error bars 1 standard deviation. * Indicates significant overall main effect ($p < .05$).

CONCLUSIONS

Findings suggest that the concept of challenge can be separated into technical and tactical, however, more research is required to find out its link to drill type. The RPC may inform coaches of whether training tasks challenge athletes to a sufficient level. This may help training design processes for more efficient learning and subsequent long-term performance in competition.

REFERENCES

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Table 1: Fixed effects from the linear mixed-effects model predicting RPC scores.

Effect	Estimate	95% CI	p
Skills vs Game-sense	-1.159	[-1.630, -0.688]	< .001
Tactical vs Technical	0.359	[0.093, 0.626]	.008
Drill type $\times$ RPC type	-0.257	[-0.791, 0.276]	.344